

"Connected could change your life forever."
— DANIEL GILBERT, author of *Stumbling on Happiness*

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Connected

The Surprising Power of Our Social Networks
and How They Shape Our Lives

When You Smile, the World Smiles with You

A strange thing happened in Tanzania in 1962. At a mission boarding school for girls near Lake Victoria in the Bukoba District, there was an epidemic of laughter. And this was not just a few schoolgirls sharing a joke. An irresistible desire to laugh broke out and spread from person to person until more than one thousand people were affected.

The affliction had an abrupt onset, and the initial bout of laughter lasted between a few minutes and a few hours in those affected. This was followed by a period of normal behavior, then typically a few relapses over the course of up to sixteen days. In what was to be a clue about the real nature of this epidemic, the victims often described feeling restless and fearful, despite their laughter.

The physicians who first investigated and reported on the outbreak—Dr. Rankin, a faculty member at Makerere University, and Dr. Philip, the medical officer of the Bukoba District—were extremely thorough. ¹ They found that each new patient had recent contact with another person suffering from the malady. They were able to observe that the incubation period between contact and onset of symptoms ranged from a few hours to a few days. Thankfully, as they intoned without irony, “no fatal cases have been reported.” Afflicted persons recovered fully.

The epidemic began on January 30, 1962, when three girls aged twelve to eighteen started laughing uncontrollably. It spread rapidly, and soon most people at the school had a serious case of the giggles. By March 18, ninety-five of the 159 pupils were affected, and the school was forced to close. The

pupils went home to their villages and towns. Ten days later, the uncontrollable laughter broke out in the village of Nshamba, fifty-five miles away, where some of the students had gone. A total of 217 people were affected. Other girls returned to their village near the Ramanshenye Girls' Middle School, and the epidemic spread to this school in mid-June. It too was forced to close when forty-eight of 154 students were stricken with uncontrollable laughter. Another outbreak occurred in the village of Kanyangereka on June 18, again when a girl went home. The outbreak started with her immediate family and spread to two nearby boys' schools, and those schools were also forced to close. After a few months, the epidemic petered out.

Rankin and Philip looked hard for biological causes for the epidemic. They performed physical examinations and lab studies on the patients, did spinal taps, examined the food supply for toxins, and ascertained that there was no prior record of a similar epidemic in the region. The villagers themselves did not know what to make of it. In Bukoba, where the illness aroused great interest, there was the "belief that the atmosphere had been poisoned as a result of the atom bomb explosions." Others described it as a kind of "spreading madness" or "*endwara yokusheka*," which means simply, "the illness of laughing."

As the villagers and the scientists investigating this outbreak realized, the epidemic was no laughing matter. It did not involve the spread of real happiness and joy—though this can happen too, albeit not in quite the same way. Rather, the outbreak was a case of epidemic hysteria, a condition that takes advantage of a deep-rooted tendency of human beings to exhibit emotional contagion. Emotions of all sorts, joyful or otherwise, can spread between pairs of people and among larger groups. Consequently, emotions have a collective and not just an individual origin. How you feel depends on how those to whom you are closely and distantly connected feel.

Our Ancestors Had Feelings

We all have emotions. And they consist of several elements. First, we usually have a conscious awareness of our emotions: when we are happy, we know it.

Second, emotions typically affect our physical state: we show how we feel on our faces, in our voices, even in our posture; given the role emotions play in social networks, these physical manifestations are especially important. Third, emotions are associated with specific neurophysiological activity; if you are shown a scary picture, the flow of blood to structures deep in your brain instantly changes. Finally, emotions are associated with visible behaviors, like laughing, crying, or shrieking. ²

Experiments have demonstrated that people can “catch” emotional states they observe in others over time frames ranging from seconds to weeks. ³ When college freshmen are randomly assigned to live with mildly depressed roommates, they become increasingly depressed over a three-month period. ⁴ Emotional contagion can even take place between strangers, after just ephemeral contact. When waiters are trained to provide “service with a smile,” their customers report feeling more satisfied, and they leave better tips. ⁵ People’s emotions and moods are affected by the emotional states of the people they interact with. Why and how does this happen?

We might consider another question first: Why aren’t emotions merely internal states? Why don’t we just have our own private feelings? Having feelings is surely evolutionarily advantageous to us. For example, the ability to feel startled is probably good for us in situations where we need to react quickly to survive. But we do not just feel startled, we show that we are startled. We jump or shriek or curse or clench, and these actions do not go unnoticed. They are copied by others.

Given the organization of early hominids into social groups, the spread of emotions served an evolutionarily adaptive purpose. ⁶ Early humans had to rely on one another for survival. Their interactions with the physical environment (weather, landscape, predators) were modulated or affected by their interactions with their social environment. Humans bonded with others in order to face the world more effectively, and mechanisms evolved to support this bonding, most obviously verbal communication but also emotional mimicry. The development of emotions in humans, the display of emotions, and the ability to read the emotions of others helped coordinate group activity by three means: facilitating interpersonal bonds, synchronizing behavior, and communicating information.

Emotions and emotional contagion probably first arose to facilitate mother-infant pair bonding and then evolved to extend to kin members and ultimately to nonkin members. Emotional contagion fosters interaction

synchrony. At the level of mother-child pairs, emotional contagion may have prompted mothers to be more attentive to and protective of their babies when their babies needed attention. Indeed, we are sadder when our family members are sad than when strangers are sad. There is an advantage in coordinating our moods with those to whom we are related.

Eventually this type of synchrony in mood or activity may have been beneficial for larger group activities, such as warding off enemies or hunting prey. If you are trying to coordinate a hunting party, it helps if members of the group are all upbeat and fired up. Conversely, if you are part of a group and someone in it appears afraid, perhaps that person has seen a predator that you have not seen. Quickly adopting his emotional state can enhance your prospects for survival. Indeed, it is thought that positive emotions may work especially well to increase group cohesiveness (“I’m happy; stay with me”) and that negative emotions may work well as communication devices (“I smell smoke; I’m scared”).

Emotions may be a quicker way to convey information about the environment and its relative safety or danger than other forms of communication, and it seems certain that emotions preceded language. What emotions lack in specificity compared to oral language, they may make up for in speed. You can tell whether your spouse is mad at you very quickly, but having her explain it to you may take a good deal more time (especially if she insists that you guess why she is mad before she tells you). You can walk through the door at home at the end of the day and immediately know whether the environment is safe or dangerous, and that is quite a trick our ancestors bequeathed us.

Of course, rapidly coordinated emotions are not always a good thing. If you come home and are in a bad mood, your partner will often detect it long before you resort to the more laborious process of explaining why you are in a bad mood. And before you have a chance to explain, she might already have caught your bad mood, which may lead to an argument and a downward spiral.

Emotional Contagion

Emotions spread from person to person because of two features of human interaction: we are biologically hardwired to mimic others outwardly, and in mimicking their outward displays, we come to adopt their inward states. If your friend feels happy, she smiles, you smile, and in the act of smiling you also come to feel happy. In bars and bedrooms, at work and on the street, everywhere people interact, we tend to synchronize our facial expressions, vocalizations, and postures unconsciously and rapidly, and as a result we also meld our emotional states.

Nowhere do we show our emotions more than on our faces. It is not difficult to explain why our facial expressions change in response to environmental stimuli or how this may be evolutionarily adaptive. Recent research, for example, has provided insight into how two facial expressions, fear and disgust, moderate our reception of sensations coming from the outside world.⁷ When we are terrified, our eyes widen and our nostrils flare to help us see and smell more of our surroundings, just as the ears of a dog perk up when it hears something interesting. Similarly, when we are disgusted, such as by an offensive odor, our noses wrinkle and our eyes narrow to reduce the impact. Air intake increases when we are afraid and decreases when we are disgusted.

Yet, facial expressions appear to have evolved not just to modify our experience of the world as individuals but as a way to communicate with others. Over time, this aspect of facial expressions probably eclipsed their original role. Such changes happen often in evolution. Feathers may have arisen merely to insulate the bodies of prehistoric reptiles, but they wound up contributing to a different and more important advantage, the ability to fly.

We developed an ability to read the facial expressions of others. Hence, we benefit when our own faces are contorted in disgust *and* by being able to notice whether others' faces are contorted in disgust. Humans have an extraordinary knack for detecting even small changes in facial expressions. This ability is localized in a particular area of the brain and can even be lost, a condition tongue-twistingly known as *prosopagnosia*. Reading the expressions of others was probably a key step on the way toward synchronizing feelings and developing the emotional empathy that underlies the process of emotional contagion.

Even as early as 1759, it was apparent to founding economist and philosopher Adam Smith that conscious thought was one way we could feel for others and hence feel like others: "Though our brother is upon the rack...

by the imagination we place ourselves in his situation, we conceive ourselves enduring all the same torments, we enter as it were into his body, and become in some measure the same person with him, and thence form some idea of his sensations, and even feel something which, though weaker in degree, is not altogether unlike them.” ⁸

However, emotions spread in ways beyond simply reading faces and thinking about the experiences of others. There is actually a more primitive, less deliberative process of emotional contagion, a kind of instinctive empathy. People imitate the facial expressions of others, then, as a direct result, they come to feel as others do. This is called *affective afference*, or the facial-feedback theory, since the path of the signals is from the muscles (of the face) to the brain, rather than the more usual, efferent pathway from the brain to the muscles. The beneficial effects of facial expressions on a person’s mood are among the reasons, for example, that telephone operators are trained to smile when they work, even though the person at the other end of the line cannot see them. This theory also explains why it helps to smile when your heart is breaking.

One biological mechanism that makes emotions (and behaviors) contagious may be the so-called *mirror neuron system* in the human brain. ⁹ Our brains practice doing actions we merely observe in others, as if we were doing them ourselves. If you’ve ever watched an intense fan at a game, you know what we are talking about—he twitches at every mistake, aching to give his own motor actions to the players on the field. When we see players run, jump, or kick, it is not only our visual cortex or even the part of our brain that thinks about what we are observing that is activated, but also the parts of our brain that would be activated if we ourselves were running, jumping, or kicking.

In one experiment related to emotional contagion, subjects listened to recordings of nonverbal vocal reactions communicating two positive emotions, such as amusement and triumph, and two negative emotions, such as fear and disgust. Investigators monitored the subjects’ brains for a response by placing them in a magnetic resonance imaging (MRI) machine. ¹⁰ The subjects were told not to react to what they heard. While subjects did not visibly respond to the sounds, the MRI results showed that hearing the cues stimulated parts of the brain that command the corresponding facial expressions. It seems we are always poised to feel what others feel and to do what others do.

Emotional Stampedes

Everyone has experience with emotional contagion: we share a joke with a friend, we feel sad when a spouse cries, we rage against city hall with our neighbors, and we hug our kids tight when they've had a bad day. Yet one often overlooked aspect of all this sharing is that emotions spread not only to our friends but to our friends' friends and beyond—even when we are not present. We are like a herd of buffalo quietly grazing on the plain until one of our neighbors starts to run. Then we start to run, and others start to run, and suddenly, mysteriously, the whole herd is barreling forward.

Epidemics of emotional states have been reported for centuries. They just have not involved laughter like the Bukoba outbreak. When emotions spread from person to person and affect large numbers of people, it is now called *mass psychogenic illness* (MPI) rather than the old-fashioned and more poetic epidemic hysteria. MPI is a specifically social phenomenon involving otherwise healthy people in a psychological cascade. Like a single startled buffalo within a herd, a single emotional reaction in one person can sometimes cause many others to feel the same thing, creating an emotional stampede.

There are two main types of MPI. In the *pure-anxiety type*, people may feel a variety of physical symptoms, including abdominal pain, headache, fainting, shortness of breath, nausea, dizziness, and so on. In the *motor type*, people may engage in hysterical dancing, pseudoseizures, and—yes—laughing, though the actual feelings underlying these behaviors are fear or anxiety. Both types of MPI thus involve the same basic psychological processes.

Historical records of such phenomena date back to at least 1374, when, in close succession to the Black Death in Europe, “dancing manias” broke out. The first such manias occurred in what is now Aachen, Germany. As described by the German medical historian J. F. C. Hecker in his 1844 book *The Epidemics of the Middle Ages*, these consisted of people who “united by one common delusion, exhibited to the public both in the streets and in the churches the following strange spectacle. They formed circles hand in hand, and appeared to have lost all control over their senses, continued dancing, regardless of the bystanders, for hours together, in wild delirium, until at length they fell to the ground in a state of exhaustion. They then complained

of extreme oppression, and groaned as if in the agonies of death.” ¹¹ These people were obviously no happier to be dancing than the African schoolgirls were to be laughing.

In a bygone era, demons and witchcraft were often seen as causes of these symptoms, but today toxic chemicals and environmental contamination are the triggers subjects typically identify. Yet, while toxins do cause some outbreaks of physical illness, they do not cause outbreaks of MPI. The source of the problem, as well as the mechanism of transmission, is psychological. Individuals afflicted in these outbreaks, and many observers, are often reluctant to ascribe the symptoms to a psychological source, however.

A relatively recent example of MPI occurred at the Warren County High School in McMinnville, Tennessee. At the time, the school had 1,825 students and 140 staff members. On November 12, 1998, a teacher believed she smelled gasoline, which caused her to complain of headache, shortness of breath, dizziness, and nausea. Seeing her response, some of her students soon developed similar symptoms. As the classroom was being emptied, other students, observing what was happening, began to report feeling unwell too. A schoolwide fire alarm was activated, and the school was evacuated. The teacher and several students were transported by ambulance to a nearby hospital, in full view of other students and teachers who were outside because of the alarm. Large numbers of police, firefighters, and emergency medical personnel from three counties responded. A total of one hundred people went to the hospital that day, and thirty-eight were admitted. Classes were canceled.

The school was closed for four days. It was inspected by the fire department, the gas company, and state officials from the Occupational Safety and Health Administration (OSHA), but no problems were identified. After the school had been deemed perfectly safe, the students and the teachers were allowed to return. Unfortunately many still smelled odors, and on November 17, seventy-one people were stricken. Ambulances were again called, and the school was evacuated and then closed.

The school's principal was fed up. In a “no more Mr. Nice Guy” move, he decided to call several government agencies, including the famed Epidemic Intelligence Service of the Centers for Disease Control (CDC). Also involved were the federal Environmental Protection Agency, the Agency for Toxic Substances and Disease Registry, the National Institute for Occupational Safety and Health, OSHA, the Tennessee Department of Health, the

Tennessee Department of Agriculture, and numerous other local emergency organizations and personnel. The investigation was extremely thorough. Aerial surveillance identified potential environmental sources of contamination; personnel explored caves in the vicinity of the school; the school's air-handling, plumbing, and structural systems were thoroughly checked; core samples were drilled from the grounds around the school; and air samples (including from the days of the outbreak) and water and waste samples were tested. The air was evaluated with an astonishing array of technology, including colorimetric tubes, flame-ionization detectors, photoionization detectors, radiation meters, and combustible-gas indicators.

Two years later, a *New England Journal of Medicine* article described the extensive examination of possible environmental causes for the illness and reported the results of the investigation by the CDC. In the end, like Rankin and Philip studying the African laughter epidemic, the investigators concluded that psychogenic factors were to blame. They found that the illness was associated with directly observing another ill person during the outbreak and with being female. ¹² The diagnosis was epidemic hysteria.

This diagnosis did not sit well with the community, and it upset many of those who had been ill, such as one twelfth grader who was quoted as saying, "They said we were crazy... . It just made me mad. When I'm sick, I don't want someone to say I'm faking. They wouldn't have taken me to the hospital, and my blood pressure wouldn't have been sky-high, if I wasn't sick." ¹³ Of course, the symptoms of those with MPI, whether laughing, dancing, fainting, or nausea, are quite real; they do not "fake" their experience in the deliberate, premeditated way that a malingerer does. The astonishing reality is that our own anxiety makes us sick, but so does the anxiety of others.

The CDC investigators also discussed why communities tended to use so many resources to try to find environmental causes for conditions that appeared to be psychogenic. The problem is that while public health professionals often suspect that an outbreak is psychogenic, they feel they have no choice but to conduct an unreasonably thorough investigation because of intense anxiety in the community. And, of course, it is very difficult, if not impossible, to definitively prove that a mysterious toxic exposure has not simply escaped detection. The CDC investigators noted the possibility of a negative community reaction to an episode labeled as psychogenic, saying, "Physicians and others are understandably reluctant to

announce that an outbreak of illness is psychogenic because of the shame and anger that the diagnosis tends to elicit.” [14](#)

An Unbearable Sweetness

Outbreaks of epidemic hysteria are not restricted to children and schools. They have been documented in adults too. One systematic review of cases of epidemic hysteria identified seventy outbreaks that occurred between 1973 and 1993 and found that 50 percent of them took place in schools, 40 percent in small towns and factories, and only 10 percent in other settings. [15](#) The outbreaks usually involved at least thirty people, and often hundreds. Most outbreaks lasted less than two weeks, but 20 percent lasted more than a month.

One of the more improbable examples was the case of the “phantom anesthetist of Mattoon.” In 1944, over a period of a few weeks during the climax of World War II, many adult residents of Mattoon, Illinois, became convinced that an “evil genius” was on the loose in their town of fifteen thousand people. This unseen person would open bedroom windows and spray victims with a “sweet-smelling” anesthetic gas that would temporarily paralyze them but, strangely, leave others in the same room unaffected. Citizens banded together to form armed patrols, but the anesthetist was never caught. The local sheriff, fearing that an innocent person might be shot, eventually ordered the posses to disband. As one investigator of this outbreak dryly noted, “The ‘gasser’ hypothesis asserts that the symptoms were produced by a gas which was sprayed on the victims by some ingenious fiend who has been able to elude the police. This explanation... is widely believed in Mattoon at present. The alternative hypothesis is that the symptoms were due to hysteria.” [16](#)

Another, more recent case occurred in 1990 among the Triborough Bridge toll employees in New York City. On February 16, workers began to complain of headaches, abdominal discomfort, dizziness, and throat and chest pain. More and more workers came down with the same symptoms over the next several days, with some of the ill workers noting what they described as a “sweetness” in the air. Symptoms were reported when workers were inside

or near a toll booth, but they would subside soon after workers left the booths. The outbreak ended on February 22, when some of the workers' superiors sat with them at the tolls. By that time, thirty-four workers had become ill enough to go to the hospital, and many others shared their symptoms. After spending hundreds of thousands of dollars searching in vain among dozens of potential culprits for a physical cause of the symptoms, it became clear to many that the illness was psychogenic. It forced 44 percent of the female workers to go to the hospital, almost twice the proportion of male workers with debilitating symptoms.

These cases share many characteristics of MPI. The symptoms tend to pop up in and spread through highly connected communities (with high network transitivity). These communities tend to be isolated and stressed. A physical culprit is seldom found. In most cases, the majority of those affected are women. It is not clear why the incidence in women and girls is higher, but it is possible that because women are inclined to discuss their symptoms, more sympathy cases result in other women. The fact that women have a more sensitive sense of smell might also play a role.

For some reason that is not well understood, smells, both real and imagined, are frequent triggers of modern outbreaks of MPI. This may have to do with the well-established connection between olfaction and emotions. Experiments have demonstrated that smell and emotion are both regulated by a part of the brain called the orbito-frontal cortex. ¹⁷ Experiments have also shown that memories evoked by smell induce stronger emotions than those evoked by verbal descriptions of the same odor. ¹⁸ Words are powerful, but one familiar whiff can jolt the mind into the past with more emotional intensity than can a signal from any other sense. This is called the *Proust phenomenon*, after the author who described a poignant memory inspired by the scent of a cookie. Smelling a perfume associated with a happy memory leads to more activity in the amygdala (a part of the brain involved in emotion and emotional memory) than seeing the bottle that the perfume comes in. ¹⁹

Paradoxically, the presence of official personnel—whether police officers, rescue workers, scientific investigators, or government officials—often worsens the epidemic, for it reinforces the belief that something serious is going on and that the situation is potentially dangerous. When these same officials attempt to provide reassurance that the situation is safe and that no cause was found, it typically generates deep suspicions among the

emotionally charged populace that a cover-up is under way, especially because the official response was previously so substantial. Paranoia can spread too, undermining the very authority that is needed to bring an end to such an episode.

The recommended treatment for MPI outbreaks focuses on social networks and recognizes that social ties are the medium for spread. The psychological guidelines for emergency workers include “providing reassurance... using a calm and authoritative approach” and “separating those who are ill from those who are not.” ²⁰ As one expert put it, “You can only stop these things by being honest... . I could get caught up in this kind of thing too, as a parent or just a person. We all could. It’s a very powerful thing, and it needs to be respected and understood. And health officials shouldn’t be so scared to call a spade a spade.” ²¹

It’s often difficult to establish why exactly these epidemics start. Just as an unfamiliar noise can trigger a cattle herd to start running, many triggers can cause emotional stampedes. However, it is usually fairly simple to identify the initial cases. For example, in the African laughing epidemic, even though the investigators could not explain why it started, they easily located the first girls to have symptoms.

It only took a few people to start *La Ola* in the stadium in Mexico City or to get passersby to stop and look up at a window in New York City, and the same is true of MPI outbreaks. When a small group of people begin acting in concert or experiencing similar, visible symptoms, the epidemic can spread along social-network ties via emotional contagion, and large groups can very quickly become emotionally synchronized.

The present obsession with nut allergies in the United States may be a case in point. The number of schools declaring themselves to be entirely “nut free” is by all accounts rising. Nuts and staples like peanut butter are prohibited from campus, and so are homemade baked goods and any foods without detailed ingredient labels. School entrances have signs admonishing visitors to wash their hands before entering to safeguard students from possible contamination.

Approximately 3.3 million Americans are allergic to nuts, and even more, 6.9 million, are allergic to seafood. However, all told, serious allergic reactions to foods cause just two thousand hospitalizations per year (out of more than thirty million hospitalizations nationwide). And, at most, only 150 people (both children and adults) die each year from food allergies. Compare

that to the fifty people who die each year from bee stings, the hundred who die from lightning strikes, and the forty-five thousand who die from motor vehicle accidents. Or compare that to the ten thousand children who are hospitalized each year for traumatic brain injuries acquired during sports, or the two thousand who drown, or the roughly thirteen hundred who die from gun accidents. Yet there are no calls to end athletics. There are likely thousands of parents who rid their cupboards of peanut butter but not guns. And more children assuredly die walking or being driven to school each year than die of nut allergies.

The question is not whether nut allergies exist, or whether they can occasionally be serious, or whether reasonable accommodations should be made for the few children who have documented serious allergies. The question is, what accounts for society's extreme response to nut allergies? Not surprisingly, the response bears many of the hallmarks of MPI. A few people have clinically documented concerns, but others who do not then copy the behaviors of those who do. Anxiety spreads from person to person to person, and a sense of proportion and the ability to be reassured are lost.

Well-intentioned efforts to reduce nut exposure actually fan the flames since they indicate to parents that nuts are a clear and present danger. This encourages more parents to worry, which fuels the epidemic. It also encourages more parents to have their kids tested, thus detecting mild and meaningless allergies to nuts. And, finally, this encourages still more avoidance of nuts, which may actually lead to a rise in true nut allergies because lack of exposure to allergens early in life is thought to contribute to the onset of allergies later. [22](#)

MPI is a pathological phenomenon, but it takes advantage of a nonpathological process that is fundamental in humans, namely, the tendency to mimic the emotional state of others. Real laughter also can be contagious and so can real happiness. But comparing epidemic hysteria to these more normal processes is like comparing the stampede of a herd to its more usual and orderly migration.

Tracking the Spread of Emotions

Measuring the subjective experience of emotions (as compared with their visible, biological, or neurological manifestations) requires asking people how they are feeling. One of the more systematic ways of doing this is known as the *experience-sampling method*. This method uses a series of alerts (such as signals sent to a beeper or cell phone) at unexpected times to prompt subjects to document their feelings, thoughts, and actions while they are experiencing them. ²³ The result is a thorough picture of the ups and downs of subjects' daily emotional lives.

One of the advantages of this method is that it allows groups of interacting people to be evaluated simultaneously in real time. For example, one team of investigators, interested in the spread of emotions within families, outfitted fifty-five families (consisting of a mother, father, and one adolescent) with beepers for one week. The participants were beeped roughly every 90 to 120 minutes between 7:30 a.m. and 9:30 p.m., and a total of 7,100 time points were observed in these 165 individuals. Various emotional states were measured, such as whether the subjects were happy or unhappy. Although the investigators could not rule out the possibility that the entire family was simultaneously exposed to one thing that made them all sad or happy at once (a confounding effect that we will discuss in greater detail in [chapter 4](#)), they did try to tease out how emotions spread within these families.

The strongest path was from daughters to both parents, while, conversely, the parents' emotional state appeared to have no effect on their daughters. Fathers' emotions affected their wives and their sons but not their daughters. This appeared to be especially true when fathers returned from work: when dad came home in a lousy mood, he soon made the whole household miserable. ²⁴

A similar method has been used to examine the transmission of emotions among teams of nurses, athletes, and even accountants. ²⁵ In such professional settings, a key question was whether one fired-up team member could improve the mood and thus the performance of his teammates. Not surprisingly, positive mood is associated with a range of team-performance-enhancing changes, including greater altruistic behavior, increased creativity, and more efficient decision making. A nice demonstration involved outfitting thirty-three professional male cricket players with pocket computers that recorded their moods four times a day during a match (which can have the insane duration of five days). There was a strong association between a player's own happiness and the happiness of his teammates, independent of

the state of the game; further, when a player's teammates were happier, the team's performance improved.

The Spread of Happiness

Despite the biological and psychological evidence for emotional mimicry, and the numerous cases of MPI arising from epidemic anxiety, until recently little was known about the precise role of social networks in the spread of emotions. Yet, the MPI cases suggest that emotions spread far and wide, flowing through social-network ties from person to person to person, and that there should be a normal analogue to this pathological phenomenon. Indeed, there can be waves of emotions in the vast fabric of human social relationships, so that people in particular locations in the social network have one emotional experience, and others elsewhere who come under different influences have a different experience altogether.

Strangely, while researchers in diverse fields, including medicine, economics, psychology, neuroscience, and evolutionary biology, have identified a broad range of stimuli of individual human happiness, they have not addressed a key (perhaps *the* key) determinant: the happiness of others. It may be obvious that our friends and family can make us happy, but before we undertook our own investigation, no one had ever explored how happiness can spread through social networks from person to person to person.

We became curious about this. We were particularly interested in determining whether the spread of emotions occurred not just between you and your friends (dyadic spread) but also between you and your friends' friends, and their friends, and beyond (hyper-dyadic spread). How far did emotions travel in the network? And were there geographic or temporal constraints on the spread?

Our first step in answering these questions was to assemble a data set that had measures of emotions and social connections over time. (We discuss that process in [chapter 4](#).) We then created a graph of the social network of happiness, as shown in plate 1. This illustration shows ties among siblings, friends, and spouses in a sample drawn from 12,067 people originally from Framingham, Massachusetts, in the year 2000, along with their levels of

happiness. No one had ever plotted such a graph before. One thousand twenty people are represented, and each node is colored on a spectrum from blue (unhappy) to yellow (happy) according to the subject's level of happiness. Looking at this image suggests two observations. First, unhappy people cluster with unhappy people in the network, and happy people cluster with happy people. Second, unhappy people seem more peripheral: they are much more likely to appear at the end of a chain of social relationships or at the edge of the network. [26](#)

Clustering of this kind in social networks can arise from a variety of processes. Happy people might choose each other as friends or be exposed to the same environments that cause them all to be happy at the same time. But our analyses allowed us to adjust for these effects. And we found that clustering is also due to the causal effect of one person's happiness on another's. Mathematical analyses of the network suggest that a person is about 15 percent more likely to be happy if a directly connected person (at one degree of separation) is happy. And the spread of happiness doesn't stop there. The happiness effect for people at two degrees of separation (the friend of a friend) is 10 percent, and for people at three degrees of separation (the friend of a friend of a friend), it is about 6 percent. At four degrees of separation, the effect peters out. Here we have our first evidence of the Three Degrees of Influence Rule. Emotions (and, as we will see later, norms and behaviors) spread in social networks from person to person to person, but they do not spread to everyone. Just as a ripple in a pond eventually fades away, so too does the ripple of an individual's happiness fade through the social network.

At first glance, these effects may not seem very significant. But compare them to the effect of having a higher income. An extra \$5,000 in 1984 dollars (which corresponds to about \$10,000 in 2009 dollars) was associated with only a 2 percent increased chance of a person being happy. So, having happy friends and relatives appears to be a more effective predictor of happiness than earning more money. And the amazing thing is that even people who are three degrees removed from you, whom you may have never met, can have a stronger impact on your personal happiness than a wad of hundreds in your pocket. Being in a particular spot in a social network, exposed to people with particular feelings, has important implications for your life.

It is well known that having more friends and relatives is much more likely to put a smile on your face than having more cash. [27](#) But past research

had never considered why friends matter so much. There are at least two possibilities. First, the existence of the social relationship itself may improve your happiness—this is a structural effect of the network on you (the second rule of social networks described in [chapter 1](#)). As we discuss in [chapter 7](#), we are hardwired to seek out social relationships, so it is not surprising that we feel pleasure or reward when we spend time with friends and family. Second, friends and relatives make us susceptible to emotional contagion, so our friends' emotional states affect our own (the third rule of social networks).

While both of these mechanisms probably contribute to people's happiness, our evidence suggests that contagion may be the more important of the two. We found that each happy friend a person has increases that person's probability of being happy by about 9 percent. Each unhappy friend decreases it by 7 percent. So if you were simply playing the averages, and you didn't know anything about the emotional state of a new person you just met, you would probably want to be friends with her. She might make you unhappy, but there is a better chance she will make you happy. This helps to explain why past researchers have found an association between happiness and the number of friends and family. But once we control for the emotional states in one's friends, we find that having more friends is not enough—having more happy friends is the key to our own emotional well-being.

This does not mean that the structure of the social network is unimportant. Amazingly, it is not just the number of dyadic ties that has an impact; the number of hyperdyadic ties also influences a person's happiness. When we measured the centrality of each person in the social network, we found that people with more friends of friends were also more likely to be happy. And, more remarkably, this was true even among people who had the same number of direct social relationships. This means that the more friends your friends have (regardless of their emotional state), the more likely you are to be happy.

One might wonder if there is a chicken-and-egg problem here. After all, it is possible to imagine that when we become happier, we tend to attract more friends, and more friends who have lots of friends. This would mean that happiness is driving the network rather than the other way around. But when we examined how the network changes over time, we found that happy people do not tend to become more central. So having a wide social circle can make you happy, but being happy does not necessarily widen your social

circle. Being located in the middle of the network leads to happiness rather than the other way around. The structure of your network and your location in it matter.

Given how important direct interaction seems to be for emotional contagion to occur, we also theorized that the effect of the happiness of your social contacts on your emotional state should depend on how near or far they are. The idea is that people who live nearby are more likely to be in contact and therefore more likely to pick up on each others' moods. Geographic distance can be used as a proxy for frequency of social interaction. In our study, about one in three people live within a mile of their closest friend, but there is a lot of variation, and some friends live thousands of miles apart. We found that when a friend who lives less than a mile away becomes happy, it can increase the probability that you are happy by 25 percent. In contrast, the happiness of a friend who lives more than a mile away has no effect. Similarly, if your spouse lives with you and he or she becomes happy, then your probability of happiness goes up, but spouses who do not live together (because they are separated) have no effect on each other. A happy sibling who lives less than a mile away increases your chance of happiness by 14 percent, but more distant siblings have no significant effect. And happy next-door neighbors also increase your chance for happiness, while neighbors who live farther away (even on the same block) have no significant effect.

All these findings suggest the importance of proximity among people whose emotions influence each other, and the impact of immediate neighbors suggests that the spread of happiness may depend as much on frequent face-to-face interaction as on deep personal connections. While in this case we are considering the spread of a dispositional state of some duration, these findings are also in keeping with the work on facial mimicry we discussed earlier.

Happiness is thus not merely a function of individual experience or choice; it is also a property of groups of people. Changes in individual happiness can ripple through social connections and create large-scale patterns in the network, giving rise to clusters of happy and unhappy individuals. Since our work was published, similar results on the spread of happiness have been observed in a sample of ten thousand rural Chinese villagers. ²⁸ Although we could not observe what causes happiness to spread, a variety of mechanisms are conceivable. Happy people may share their good fortune (e.g., by being pragmatically helpful or financially generous to

others), change their behavior toward others (e.g., by being nicer or less hostile), or merely exude an emotion that is contagious. Being surrounded by happy people might have beneficial biological effects too. But whatever the mechanism, it seems clear that we need to change the way we think about happiness and other emotions.

Life on the Hedonic Treadmill

We all know people who are hedonists; they can never get enough of the good life. In fact, lasting happiness is difficult to achieve because people are on a “hedonic treadmill.” Although a change in a person’s circumstances may cause him to be happier (e.g., finding a partner, winning the lottery) or sadder (e.g., losing a job, becoming paralyzed), a broad body of research has shown that people tend to return to their previous level of happiness after such events. ²⁹ In fact, studies of lottery winners and spinal cord injury patients reveal that after a year or two, they are often no more happy or sad than the rest of us. Our surprise at this observation stems in part from our inability to anticipate that some things will not change. Lottery winners still have annoying relatives, and paralyzed patients can still fall in love. As psychologist Daniel Gilbert has shown, we tend to focus on only the most salient part of a situation when we are thinking about things that might befall us. ³⁰ Moreover, we overlook our ability to adapt to circumstances. So, a person trying to become happier is like someone walking up a downward-moving escalator. Although the effort to climb up and become happier is helpful, it is counteracted by the process of adaptation that forces one back to one’s original state.

Many people try to overcome this problem by intentionally engaging in activities to improve their happiness. We might change our behavior by exercising regularly or by trying to be kind to others or even by avoiding a long commute (which has been shown to be particularly deleterious to happiness). We might change our attitude by pausing to count our blessings or thinking about experiences in the most positive light (as Tibetan monks do). We might also devote effort to causes we find meaningful or strive to achieve important personal goals. Indeed, there is reason to suspect that a

sustained effort to engage in such happiness-producing activities might help us progress up the downward-moving escalator.

But in spite of these efforts, each of us tends to stay put in a particular long-term disposition; we appear to have a set point for personal happiness that is not easy to change. In fact, like other personality traits, personal happiness appears to be strongly influenced by our genes. Studies of identical and fraternal twins show that identical twins are significantly more likely to exhibit the same level of happiness than are fraternal twins or other siblings. Behavior geneticists have used these studies to estimate just how much genes matter, and their best guess is that long-term happiness depends 50 percent on a person's genetic set point, 10 percent on their circumstances (e.g., where they live, how rich they are, how healthy they are), and 40 percent on what they choose to think and do. ³¹ What we experience in life can, of course, change our moods for a period of time, but in most cases these changes are transitory.

What about the network spread of happiness? Does it obey this constraint, only making us happy for a while? Does the effect of having a friend become happy tend to wear off? In our study, we found that a person is 45 percent more likely to be happy if a friend became happy in the previous six months. In contrast, the effect is only 35 percent for friends who became happy within the previous year, and it disappears after longer periods of time. So, our friends' happiness does have an effect on us, but it only lasts for about a year. Just as lottery winners get used to their newfound wealth, we get used to our friends being happy. But if different friends get happy at different points in time, they might give us a periodic boost, helping us to stay above our natural level of happiness.

Alone in the Crowd

If happiness can spread, at least for a while, what about other emotions? One feeling that directly concerns our social network is loneliness. In some sense, loneliness is the opposite of connection—it is the feeling of being disconnected. Work by psychologist John Cacioppo has shown that loneliness is a complex set of feelings experienced by people whose core

needs for intimacy and social connection are not met. ³² This often motivates most (but not all) people to redress their situation, suggesting that the function of loneliness is to promote reconnection (we will discuss the evolutionary purpose of loneliness in [chapter 7](#)).

Psychologists have identified the way that feelings of loneliness fit in with a broad set of other feelings and states, including self-esteem, anxiety, anger, sadness, optimism, and shyness. Psychological research suggests that feelings of loneliness occur when there is a discrepancy between our desire for connection to others and the actual connections we have. This research has focused on the subjective perception of being alone, but feeling lonely is not the same thing. While studies have shown, unsurprisingly, that having a good friend can decrease loneliness, what has not previously been examined is the effect of the whole social network on our tendency to feel lonely even in a crowd.

Using the same network in which we studied happiness, we examined whether being alone was associated with feeling lonely and whether such feelings could spread. ³³ We found that real-world social connections do have an effect on how we feel. People with more friends are less likely to experience loneliness. Each extra friend reduces by about two days the number of days we feel lonely each year. Since on average (in our data) people feel lonely forty-eight days per year, having a couple of extra friends makes you about 10 percent less lonely than other people. Interestingly, the number of family members has no effect at all. It is not clear why this is the case. Possibly, people in small families know they have a greater responsibility to spend time with one another since there are fewer people to take turns visiting. Or perhaps people in large families primarily feel close to a smaller core of their family, limiting the influence of additional connections. Regardless of the mechanism, it is clear that feelings of loneliness are much more closely tied to our networks of optional social connections than to those handed to us at birth.

Loneliness can actually shape the social network. People who feel lonely all the time will lose about 8 percent of their friends, on average, over two to four years. Lonely people tend to attract fewer friends, but they also tend to name fewer people as friends as well. What this means is that loneliness is both a cause and a consequence of becoming disconnected. Emotions and networks can reinforce each other and create a rich-get-richer cycle that rewards those with the most friends. People with few friends are more likely

to become lonely, and this feeling then makes it less likely that they will attract or try to form new social ties.

Our study suggests that physical proximity matters as much for loneliness as it does for happiness. Friends and family who live nearby see each other more often, which should help decrease the likelihood that they feel lonely, but it also makes them more susceptible to one another's feelings. For example, if a nearby friend has ten extra lonely days a year, it will increase the number of lonely days you experience by about three. If this person is a close friend, then the effect is stronger, and you'll experience four extra days of loneliness. Loneliness also spreads between next-door neighbors, with ten extra days of loneliness leading to two extra days for the person on the other side of the fence. But neighbors and friends who live more than a mile away do not make each other lonely.

Spouses who live together can affect each other too, but the result is less dramatic. For every ten extra days a person is lonely, his or her spouse will be lonely for just one extra day. And siblings do not appear to affect one another at all (even the ones who live nearby); this provides additional evidence that loneliness is about our relationships to people with whom we choose to connect rather than the relationships we have inherited.

Looking beyond these direct connections, we found that loneliness spreads three degrees, just like happiness. A person's loneliness depends not only on his friends' loneliness, but also on his friends' friends' and his friends' friends' friends' loneliness. The full network shows that you are about 52 percent more likely to be lonely if a person you are directly connected to (at one degree of separation) is lonely. The effect for people at two degrees of separation is 25 percent, and for people at three degrees of separation, it is about 15 percent. At four degrees of separation the effect disappears, in keeping with the Three Degrees of Influence Rule.

Finally, we observed an extraordinary pattern at the edge of the social network. At the periphery, people have fewer friends; this makes them lonely, but this also tends to drive them to cut the few ties that they have left. But before they do, they may infect their friends with the same feeling of loneliness, starting the cycle anew. These reinforcing effects mean that our social fabric can fray at the edges, like a strand of yarn that comes loose from the sleeve of a sweater. If we are concerned about combating the feeling of loneliness in our society, we should aggressively target the people at the periphery with interventions to repair their social networks. By helping them,

we can create a protective barrier against loneliness that will keep the whole network from unraveling.

Feeling in Love

The psychology of emotions such as happiness and loneliness sheds light on the formation and dissolution of ties in social networks. In fact, human sensibilities such as anger, sadness, grief, and love all operate in the service of social ties. One can be angry at nature or saddened by a forest fire or love a pet, but these emotions have their origin and find their fullest expression in the anger, sadness, or love one feels in the setting of interpersonal relationships.

People the world over have different ideas, beliefs, and opinions—different thoughts—but they have very similar, if not identical, feelings. And they have similar responses to feelings in others, preferring happy friends to depressed ones, kind friends to mean ones, and loving friends to violent ones. A whole range of emotions can spread, from anger and hatred to anxiety and fear to happiness and loneliness. But there is one emotion central to human experience that we have not yet considered and that is key to understanding social connection: love.

The psychology of love and affection is obviously crucial to an understanding of the formation of social ties between people. As anthropologist Helen Fisher has argued, the sensibility of being in love may be broken down into lust, love, and attachment, all of which likely served evolutionary purposes. ³⁴ The feeling of lust has the obvious goal of driving reproduction—with almost any partner. The feeling of romantic love is something different, of course, and tends to be focused on a particular partner, or at least one partner at a time. From an evolutionary perspective, this allows the individual to conserve precious resources and not waste them in the pursuit of several objects of affection. The feeling of attachment, and the secure tie to another person that it represents, may have evolved to allow parents to jointly care for their young, which also has evolutionary advantages.

In [chapter 7](#), we will discuss the role of natural selection in human social

networks in more detail, but before we get there it is important to think about the implications of our deepest connections. Aside from the evolutionary advantages and disadvantages, feelings of lust, love, and attachment carry enormous implications for the way we connect to others. The object of one's affection becomes the "center of one's universe," around which all else revolves. People experience intrusive thoughts about their beloved, aggrandize their beloved, are energized by their beloved, and are obviously deeply connected to their beloved. We usually experience such romantic love with just one person at a time. So romantic love does not determine the general organization of social networks. After all, we do not love everyone we know. And the love we have for our parents, children, siblings, and other relations is a different kind of feeling. Yet, as we will see in the next chapter, being in love is a key mechanism by which certain important social ties are formed, and it is therefore highly relevant to the origin—and function—of social networks.